Marine Detect: AI-Powered Marine Species Detection

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***Abstract*—This project explores the use of AI-powered object detection, specifically leveraging YOLOv8, to automate the identification of marine species from underwater imagery and video. Motivated by the pressing need for scalable, real-time ecological monitoring, the system processes input media through a structured inference workflow and outputs annotated detections in YOLOv8 format. The architecture mimics cognitive memory systems to manage classifications, thresholds, and downstream outputs for researchers and databases. This approach enables efficient, high throughput monitoring of ocean life, addressing key challenges such as habitat loss and overfishing, and paving the way for more informed conservation strategies.**

***Keywords*—inferencing, marine, artificial intelligence, model, datasets, ecological monitoring,**

I. Introduction

The monitoring of marine biodiversity has become increasingly critical in light of accelerating climate change, habitat degradation, and overfishing. Video-based observation techniques have grown in popularity for assessing underwater ecosystems due to their non-intrusive nature and ability to capture continuous, high-resolution footage of marine species in their natural habitats [1]. However, analyzing such video data manually remains a major bottleneck—raising the central question: How can computer vision and machine learning be leveraged to automate marine species detection at scale and with high accuracy?

This question is particularly pressing in contexts where large volumes of underwater footage must be processed for ecological monitoring, impact assessments, or conservation enforcement. Traditional object detection models have demonstrated strong performance in terrestrial environments [2], yet their performance degrades under the unique visual conditions of underwater scenes, including poor lighting, turbidity, and high inter-class visual similarity. This creates a fundamental problem: state-of-the-art models often require adaptation, fine-tuning, and domain-specific datasets to be effective underwater.

Efforts to address this problem have included transfer learning approaches, synthetic data augmentation, and domain adaptation [3], but many existing pipelines lack accessibility and modularity, limiting their usability by non-expert users such as marine ecologists. The lack of large, labeled underwater datasets also continues to hinder progress. To support reproducible research in this domain, the Marine-Detect framework [4] was introduced as an open-source pipeline designed to simplify the application of deep learning to marine species detection. It incorporates recent model architectures such as YOLOv8 [5], supports model training and inference, and allows researchers to plug in domain-specific datasets.

Marine-Detect has been evaluated on datasets such as the BRUVS (Baited Remote Underwater Video Stations) and Fish4Knowledge collections, which contain labeled footage of multiple species in diverse underwater conditions. These datasets provide a benchmark for assessing model generalization and robustness across habitats and camera setups [6], [7].

In this paper, we examine the design and functionality of the Marine-Detect pipeline, assess its performance on multiple underwater datasets, and explore its extensibility to support real-time applications, fine-tuning with limited labeled data, and integration with active learning loops. By combining recent advances in object detection with a domain-oriented pipeline, we aim to offer a practical and extensible tool for advancing marine ecological monitoring through artificial intelligence.

II. Why We Need AI In Oceans?

The ocean, covering over 70% of Earth's surface, is home to nearly 80% of all life forms, yet remains vastly under-monitored and under-understood. As global environmental challenges intensify, the need for effective tools to protect marine ecosystems becomes increasingly urgent. Artificial Intelligence (AI) offers a transformative opportunity to bridge the vast data gaps in ocean monitoring and management.

Current methods for monitoring marine biodiversity include taking photographs and videos of sea life, classifying images, and conducting field surveys. One thing all of these examples have in common is that they are human led. For instance, scientists send scuba divers with technologies like remotely operated vehicles to take photos of the ocean floor. These images and videos are later analyzed to identify species, which are used to assess the overall health of the marine ecosystem [8]. While these traditional approaches are important, they are time-consuming and costly. They do not allow for real-time monitoring and are susceptible to human error during data collection and interpretation, which can affect reliability.

Artificial intelligence tools such as Marine-Detect address these challenges by using computer visions and machine learning to automatically detect, classify, and monitor marine species with minimal human interaction. This model allows for 24/7 monitoring of coral reefs across larger areas, which expands the scale of the monitoring. Also, the model helps

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eliminate human error which improves the accuracy of species classification. This enhances our ability to monitor the effects of environmental stressors, such as coral bleaching, pollution, and overfishing, on marine life and coral reefs in real time.

III. What Is Marine Detect?

Marine-Detect AI is an open-source marine biodiversity detection system developed through a partnership between Orange Business and Tēnaka, within the framework of the Let's Revive project [4]. The project seeks to enhance marine life and more specifically coral reef restoration by streamlining ecosystem monitoring. Tēnaka emphasizes monthly impact reporting through its Tēnaka Science initiative, and to support that mission, the Marine-Detect system was deployed via underwater research stations equipped with cameras. These stations autonomously capture image and video data in coral reef environments and transmit processed results to marine scientists in near-real time, thereby reducing manual labor and enabling experts to focus more on ecological restoration [4].

A. Model Constituents

Marine-Detect includes two dedicated object detection models designed to identify and quantify a variety of marine species: the Fish and Invertebrates Model, and the Megafauna Model. This separation is based on both ecological and technical considerations. The Fish and Invertebrates Model focuses on common reef-dwelling organisms such as Butterfly Fish (Chaetodontidae), Groupers (Serranidae), Moray Eels (Muraenidae), and invertebrates like Giant Clams, Sea Cucumbers, and Lobsters. The Megafauna Model, in contrast, is trained to recognize larger and rarer species such as Sharks, Sea Turtles, and Rays [4].

These species were selected based on their roles as bio-indicators, which are organisms whose presence or absence reflects the health of coral reef ecosystems. The current implementation focuses on species typical to the Indo-Pacific and Malaysian marine regions. The decision to use two separate models stems in part from practical considerations: many public datasets already contain annotations specific to either fish/invertebrates or megafauna. A unified model would have required extensive re-annotation across domains, whereas the dual-model strategy preserves existing labeled data and boosts detection accuracy within each category [4].

B. Model Implementation

Both models are implemented using YOLOv8 the latest evolution of the YOLO object detection architecture, developed by Ultralytics [15]. YOLOv8 is an anchor-free, convolutional neural network optimized for speed and accuracy, making it particularly well-suited for resource-constrained environments such as underwater embedded systems [15]. The models output bounding boxes and confidence scores for each detection, and are trained using the YOLO annotation format, which stores image-specific object coordinates and class identifiers in lightweight .txt files.

C. Employed Datasets

Marine-Detect's training datasets are composed of approximately 90% publicly available annotated images and 10% Tēnaka-sourced data, including both stills and frames extracted from underwater video. The dataset splits for both the Fish and Invertebrates and Megafauna models are summarized in Table I.

Model	Training + Validation Sets	Test Set
FishInv	12,243 images (80%, 20%)	499 images
MegaFauna	8,130 images (80%, 20%)	253 images

Fig. 1. *A breakdown of all training and validation sets.*

In some cases, datasets required manual labeling, while others were already annotated [4]. This hybrid dataset strategy not only enhances generalization across diverse underwater scenes but also aligns the models with real-world field deployments across coral habitats.

In summary, Marine-Detect AI serves as a robust and scalable ecological sensing tool, transforming raw visual data into structured biological insight. Its modular architecture, species-specific design, and real-time AI integration position it as a promising contributor to the future of marine conservation governance.

IV. Cognitive Systems Perspective

Marine-Detect AI operates not merely as a tool of visual classification but as an instantiation of modular, bounded cognition within a socio-technical infrastructure for marine governance. Its behavior, while narrow in scope, can be analytically mapped to the core architectural principles of the Common Model of Cognition (CMC), as outlined by Laird, Lebiere, and Rosenbloom [13]. The CMC is defined not by specific implementation (e.g., ACT-R or Soar), but by a set of abstract assumptions: task-independence of modules, parallel perceptual and procedural processing, symbolic memory structures, and sequential cognitive cycles that support bounded, rather than optimal, rationality [13].

A. Mapping Marine-Detect to the Common Model of Cognition

Marine-Detect's perception module is instantiated through YOLOv8-based object detection. Incoming visual data–captured by underwater cameras–is parsed into bounding boxes and class identifiers. In CMC terms, this represents perceptual processing that yields symbol structures with associated metadata in specific working memory buffers [13]. The system performs these perceptual operations in parallel with other tasks such as confidence scoring and output streaming, reflecting the model's support for parallel processing both across and within modules [13].

B. Memory, Motor, and the Cognitive Cycle

Once parsed, these perceptual symbols are loaded into a working memory buffer, which interfaces with external systems–research dashboards, databases, or marine scientists–enabling short-term retention and rapid access. This aligns with the CMC's conception of global communication through a short-term working memory that connects perceptual, cognitive, and motor modules [13]. The contents of this buffer–i.e., species labels and bounding coordinates–are manipulated by the system's procedural logic to determine whether detections are valid, whether further processing is needed, or whether alerts should be issued.

Long-term memory in Marine-Detect is split across declarative and procedural layers, though both are encoded within the neural architecture of the YOLOv8 model. The declarative component is reflected in the training data and label space: factual knowledge about which species exist, their appearance, and bounding definitions. The procedural component resides in the learned model weights, which determine how inputs are processed and which actions (e.g., labeling, saving, reporting) are taken. This distinction echoes Laird's description of symbol structures in declarative memory and rule-like conditions in procedural memory, both of which influence working memory and system control [13].

Despite its lack of autonomous goal-setting, Marine-Detect participates in sequential action selection via a constrained cognitive cycle. When an image is received, it is processed, classified, and forwarded to action interfaces. Each operation unfolds as part of a cycle that mimics the architectural rhythm of cognition, albeit without human-scale timescales or top-down attentional modulation. The system's logic is not optimized globally but driven by locally scoped, bounded decision loops–a hallmark of the CMC's commitment to bounded rationality and modular sequencing [13].

Crucially, Marine-Detect's symbolic outputs–species labels, positional metadata, confidence scores–are transmitted outward, supporting motor-like action in the form of report generation, data visualization, and conservation response. While it does not actuate physical devices, this mechanism satisfies the model's definition of motor control: the translation of internal relational structures into external outputs that affect the environment [13]. Taken together, Marine-Detect embodies all five architectural components of the Common Model of Cognition can be summarized in Figure 1.

What emerges is not artificial general intelligence, but a cognitive architecture in miniature: bounded, modular, symbolically structured, and environmentally situated. As Laird et al. emphasize, "complex behavior arises from a sequence of independent cognitive cycles that operate in their local context" [13]. Marine-Detect's detection pipeline, though narrow, reflects this sequencing: it perceives, categorizes, and acts–symbolically, cyclically, and under the constraints of real-world ecology.

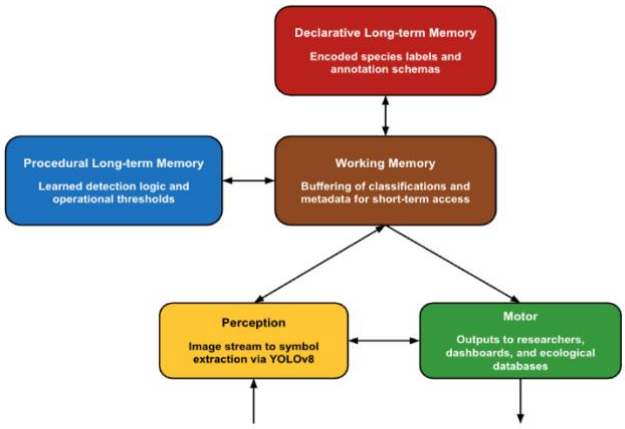


Fig. 2. *Marine-Detect AI in the context of the structure of the Standard Model of Cognition.*

C. Situated Cognition and Epistemic Infrastructure

Moreover, Marine-Detect's architecture does not operate in a cognitive or technical vacuum. It is embedded within a broader epistemic infrastructure–a sociotechnical system that governs what counts as knowledge, how that knowledge is produced, and who it is for [14]. This infrastructure includes datasets curated from specific geographic regions, species selected as taxonomically significant, and annotation protocols that prioritize certain visual features over others. The system's symbolic outputs are not neutral reflections of marine reality; they are conditioned by design choices, institutional expectations, and ecological priorities. In this sense, Marine-Detect AI does not merely model cognition in the abstract; it performs cognition within a layered ecosystem of political, ecological, and scientific meaning-making. What it detects–and what it fails to detect–shapes what is legible to conservation systems, and thereby what becomes actionable. As such, Marine-Detect is not only a cognitive architecture in the Standard Model sense, but also an epistemic actor that helps define the contours of environmental governance itself.

V. Detailing the Marine Detect System Approach

A. The Model Architecture

The Marine Detect system is built upon a robust foundation of established deep learning technologies. It leverages the PyTorch framework, known for its flexibility and extensive support for GPU acceleration via NVIDIA CUDA. This combination is crucial for handling the computationally intensive tasks involved in processing large volumes of visual data typical in marine surveys. At the core of the detection process are YOLOv8 models, specifically trained or fine-tuned for identifying marine life ("FishInv/MegaFauna" likely refers to models specialized for fish inventory and marine megafauna detection). YOLO models are renowned for their speed and efficiency in real-time object detection, making them well-suited for analyzing potentially long video sequences or large image datasets.

B. Key Tehcnical Approaches Used

Marine Detect implements several key strategies to enhance detection accuracy and usability:

Multi-model Ensemble Inference: Instead of relying on a single model, the system employs an ensemble approach. This involves using multiple YOLOv8 models, potentially trained on different datasets or with varying parameters. By combining the predictions from these diverse models, the system can achieve higher accuracy, better generalization across different underwater conditions (e.g., varying visibility, lighting), and improved robustness in identifying a wider range of species or objects compared to any single model. Considered alternatives like dynamic model selection or weighting based on input characteristics might offer future refinement but the ensemble method provides a solid baseline.

Unified Visualization: A critical aspect of the system is its ability to present the detection results clearly. After aggregating predictions, the system overlays bounding boxes and potentially label directly onto the original images or video frames. This unified visualization allows researchers and analysts to quickly verify detections and understand the spatial distribution of marine life within the analyzed media.

C. *Processing Workflow*

The system follows a structured workflow for both image and video inputs:

1. Initialization: The process begins by loading the pre-trained YOLOv8 ensemble models (e.g., Fishinv, MegaFauna).
2. Input Determination: The system checks whether the input is a single image or a video file.
3. Inference:
   - Images: Each image is processed by ensemble models. Detections are generated based on predefined confidence threshold values, filtering out low-probability results.
   - Videos: Each frame of the video is processed similarly, running the models and applying confidence thresholds.
4. Results Aggregation (combine_results): Predictions from the ensemble models (for images) or across frames/models (for videos) are aggregated. This step likely involves techniques like non-max suppression (NMS) to merge overlapping bounding boxes for the same object and consolidate the findings.
5. Post-processing & Visualization:
   - Images: Image orientation is checked and corrected using EXIF metadata, ensuring annotations are applied correctly regardless of camera orientation during capture. The aggregated detections are then overlaid onto the corrected image.
   - Videos: The aggregated predictions are drawn onto each corresponding video frame.
6. Output Generation:
   - Images: The final annotated image is saved.
   - Videos: Each annotated frame is written sequentially to an output video file using libraries like OpenCV (cv2.VideoWriter).

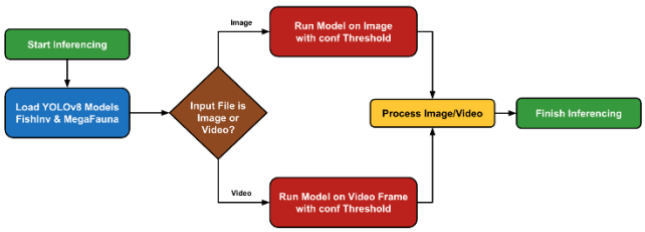


Fig. 3. *A graphical illustration of the entire flow of the model inferencing.*

This automated pipeline, powered by ensemble deep learning models and GPU acceleration, significantly reduces the manual effort required for analyzing marine visual data. It provides a scalable and efficient tool for generating valuable insights into species presence, abundance, and behavior, directly supporting conservation efforts and enhancing our understanding of critical ocean ecosystems.

D. *Results from the Workflow and Script Execution*

Upon successful execution of the inference workflow, the system produces annotated outputs in the YOLOv8 format. These results represent the core output of our detection pipeline and demonstrate the model's ability to identify and classify marine species in both image and video input streams.

Once the input is processed whether a single image or frames extracted from a video, the YOLOv8 model applies trained detection logic to assign bounding boxes and class labels to detected fish or marine objects.

```
image 1/1 /home/sourajitk/cmpsc-497-spring-2025-final-project-environment-1/marine-detect/notebooks/datasets/FishInvSplit/test/images/00125.jpg: 384x640 (no detections), 336.3ms
Speed: 4.0ms preprocess, 336.3ms inference, 4.2ms postprocess per image at shape (1, 3, 384, 640)
Results saved to output/fishinv_results
```

Fig. 4. A sample run of our script to test the YOLOv8 models with datasets.

Each annotated result includes:

1. Bounding box coordinates around detected species.
2. Class labels mapped to predefined marine categories.
3. Confidence scores indicating the model's certainty for each detection.

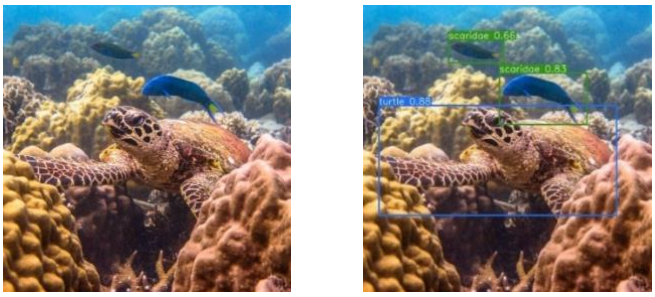


Fig. 5. *After running the model, the outputted image or video will have bounding boxes annotations to visually indicate the location and extent of the detected marine species within the original data.*

The annotations are stored in .txt files, aligned with YOLOv8's structured output schema, where each line represents a detection instance. These files are saved alongside their corresponding images or frame snapshots, creating a directly interpretable dataset for downstream tasks.

This output format enables seamless integration with visualization tools, further model training, or ecological database ingestion. Additionally, the structured results provide a consistent foundation for researchers to evaluate model performance, validate detections, and iterate improvements in species classification accuracy.

VI. Evaluation of Marine-Detect

A. *Evaluation Methodology*

To assess the performance and reliability of the Marine-Detect model under various environmental conditions, we conducted comprehensive testing using controlled image alterations. This evaluation aimed to systematically determine the model's robustness across different types of image degradation commonly encountered in underwater photography.

Our evaluation consisted of applying multiple types of controlled image degradations to our test dataset containing 344 images. These modifications included:

- Gaussian blur with kernels sizes of 3×3, 5×5, and 7×7
- Noise addition with standard deviations of 10, 20, and 30
- Brightness adjustments (dimming and brightening)
- Contrast modifications (0.5× and 1.5×)
- Color filters included blue tint, grayscale, and sepia

- JPEG compression at quality levels of 10, 30, and 50

Each alteration type was applied independently to the baseline test set to enable direct performance comparisons across different environmental conditions.

B. Experimental Setup

To ensure reproducibility and efficiency in our evaluation process, we developed three custom Python scripts:

1. Image Alteration Script (*image_alteration.py*): Automated the application of various image transformations, organizing modified images into structured directories for subsequent processing.
2. Testing Script (*test_fishinv.py*): Executed the Marine-Detect model on each dataset variant, generating annotated images with bounding boxes and comprehensive detection reports including overall and species-specific statistics.
3. Analysis Script (*summarize_detection.py*): Aggregated results across all test runs, producing CSV files for statistical analysis and performance comparison.

C. Results

1. Overall Performance Analysis:

The baseline detection rate was 87.79% (302 of 344 images with successful detections). Performance varied significantly across different image alterations:

Most challenging conditions:

- JPEG compression (Q=10): 62.5% detection rate
- Noise (σ=30): 65.12% detection rate
- Noise (σ=20): 79.07% detection rate

Most resilient conditions:

- High contrast (1.5×): 88.95% detection rate
- Low contrast (0.5×): 88.08% detection rate
- Brightness adjustments: >88% retention rate

2. Species-Specific Analysis:

Detection performance varied considerably across different marine species categories:

Crown-of-thorns starfish demonstrated exceptional robustness, maintaining 24-26 consistent detections across all conditions. In contrast, the broad "fish" category exhibited significant sensitivity to image degradation:

- Original: 184 detections
- Severe compression (Q=10): 62 detections (66% reduction)
- Heavy noise (σ=30): 74 detections (60% reduction)

Report	Count Category			
	With detections	No detections	Total detections	Detection rate
ORIGINAL	302	42	676	0.8779
3×3	302	42	652	0.8779
7×7	295	49	597	0.8576
dimmed	305	39	667	0.8866
brightened	304	40	647	0.8837
Q=10	215	129	377	0.625
Q=50	302	42	620	0.8779
σ=10	296	48	566	0.8605
σ=30	224	120	327	0.6512
sepia	303	41	652	0.8808
blue	306	38	641	0.8895
grayscale	304	40	678	0.8837
0.5×	291	53	514	0.8459
1.5×	298	46	587	0.8663

Fig. 6. *Species detection counts by alteration types*

3. Sea Cucmber Detection Anomaly

A noteworthy finding emergedd in sea cucumber detection pattens. The count increased substantially with noise addition:

Originally, we had 2 detections following with noise:

- Noise σ = 10: 4 detections
- Noise σ = 20: 13 detections
- Noise σ = 30: 29 detections

This suggests the model may be misclassifying noise artifacts as sea cucumber features. Potential explanations include: First, textural similarity between sea cucumber surface patterns and noise artifacts. Second, shape ambiguity wheere noise creates visual patterns resembling sea cucumber morphology.

Lastly, Background interaction effects where noise on sandy substrates creates false texture patterns.

Fig. 7. *Detection performance across image alterations*

Species	Image Alteration Type													
	ORIGINAL	3×3	7×7	dimmed	brightened	Q=10	Q=50	σ=10	σ=30	sepia	blue	grayscale	0.5×	1.5×
bolbometopon muricatum	20	19	16	20	20	11	19	19	11	16	22	20	20	20
chaetodontidae	55	52	49	57	55	48	52	49	36	43	55	44	54	55
cheilinus undulatus	5	5	4	6	4	5	5	5	9	4	6	5	6	4
cromileptes altivelis	1	1	2	1	1	1	1	1	1	1	3	1	1	1
crown of thorns	26	26	26	26	26	25	26	26	24	24	26	26	26	26
fish	184	175	146	181	175	62	158	132	74	115	207	170	173	174
giant clam	96	94	91	92	83	67	86	76	22	65	84	71	87	82
haemulidae	22	22	19	22	22	19	23	21	22	22	22	21	22	22
lobster	0	0	0	0	0	0	0	0	0	1	0	0	0	0
lutjanidae	82	84	81	81	83	23	75	84	39	66	81	68	81	80
muraenidae	63	56	48	65	62	44	71	56	26	74	68	60	66	63
scaridae	13	13	24	9	12	17	10	7	3	6	10	8	9	12
sea cucumber	2	3	2	2	2	2	2	4	29	2	2	4	2	2
serranidae	46	42	34	46	45	11	35	40	23	30	39	37	46	44
urchin	61	60	55	59	57	42	57	46	8	45	53	52	49	56

4. Detection Performance Visualization

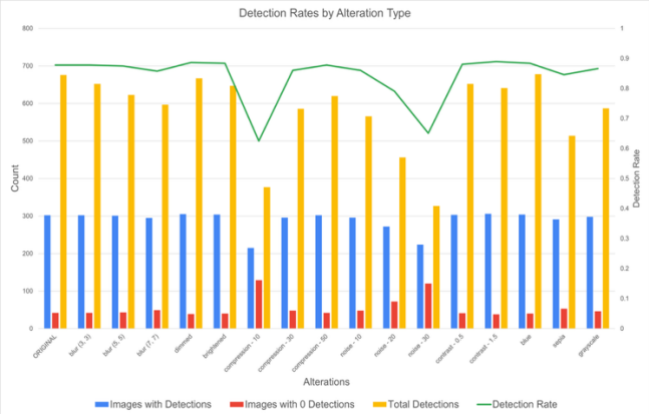


Fig. 8. *Detection rates by alteration type, showing the impact of various image modifications on model performance. Blue bars indicate images with successful detections, red bars show images with no detections, yellow bars represent total detections, and the green line shows detection rate percentage.*

The visualization in Fig. 8 clearly demonstrates the model's resilience to minor image alterations while highlighting critical failure points under extreme conditions. The detection rate line illustrates how performance degrades progressively with increasing severity of alterations, particularly for compression and noise. Notable performance degradation occurs under severe compression (Q=10) and noise (σ=30) conditions.

D. Discussion

1. Performance Characteristics

The evaluation of the Marine-Detect model revealed several important characteristics that inform its deployment and future optimization. Notably, the model demonstrates a strong reliance on shape and texture features rather than color, as its performance remains largely unaffected by color alterations but degrades with structural distortions. This feature dependency highlights its robustness to lighting variability, which is common in underwater environments. However, the model exhibits significant vulnerability to JPEG compression artifacts, particularly at lower quality settings, suggesting architectural sensitivity that could impact real-time applications where compression is often necessary. Additionally, the model's performance varies across species categories, indicating differing levels of feature complexity in the training data. This taxonomic stability suggests that broader or more diverse groups may benefit from additional fine-tuning to ensure consistent detection accuracy across classifications.

2. False Positive Detection

The sea cucumber anomaly represents a significant limitation in the model's discrimination ability. Under severe image degradation, the system generates false positive detections, mistaking noise patterns for biological features. This finding has important implications for real-world deployment where environmental conditions may introduce similar artifacts.

The results demonstrate that while the Marine-Detect system performs well under optimal conditions, its reliability decreases substantially with image quality degradation. These findings highlight the need for additional training on compressed and noisy imagery to improve robustness in practical underwater applications.

VII. Limitations To Our Approach

There are several limitations to the Marine Detect model, including environmental limitations, computational limitations, and ethical limitations. Environmental factors can significantly affect the accuracy of the model. For example, conditions underwater such as poor water clarity, poor lighting, and bad camera placement can lower the model's ability to detect and classify marine species. Motion blur caused by ocean currents, other movements, or poorly stabilized footage can distort images. These distortions lead to inconsistencies with the model, as it has a hard time classifying marine life. As a result, some poorly pictured species may be misidentified or missed altogether, reducing the reliability of the model.

Computational factors can also significantly affect the accuracy of the model. Marine Detect performs well under standard testing conditions but struggles with high levels of distortion and noise. Our dataset, which is approximately 90% public, may lack sufficient representation of rarer species, which results in the model being more biased towards certain species. Bender et al.'s "On the Dangers of Stochastic Parrots," details the implications of a biased dataset. Models that are trained on pre-existing biases increase the risk of amplifying existing social – and in this case, biological – biases [12]. By having extremely vague datasets, like "fish," our model underrepresents marine life while reinforcing social biases that all fish are similar.

The model also prioritizes shape and texture during detection, which aligns with the behavior of most neural networks. However, this also means that the model is vulnerable to changes in image quality. Training on images with lower resolution, greater compression, or added blur could improve the model. Additionally, adding more detailed classifications to the model would help identify more species.

VIII. Ethical Issues

Marine-Detect, like all artificial intelligence systems, is not neutral. It reflects and reproduces particular worldviews, priorities, and systems of valuation–those of its creators, of the datasets used to train it, and of the historical structures in which these technologies are embedded. As Langdon Winner reminds us, artifacts have politics [9], and this holds no less for AI systems that surveil and interpret coral ecosystems. The very act of determining which marine life is "worth" monitoring–where some organisms, such as the moray eel, receive higher classification granularity than others, like generic "fish"–demonstrates an embedded hierarchy of significance. This hierarchy is not biologically objective; it is constructed. And this construction reflects a broader epistemological violence–one that prioritizes visibility, legibility, and utility according to human-centered and colonial logics.

This logic is not incidental. It draws from the onto-epistemological structure that Sylvia Wynter critiques through her notion of the "Man" genre of the Human [10]. "Man1," defined through colonial, Western, white, and rationalist ideals, serves as the overrepresented archetype of the human. AI systems trained under such regimes likewise overrepresent certain categories of life–the photogenic, the endangered, the

economically relevant–and disregard the vast multiplicities that do not perform value in the ways these systems are trained to recognize. “Fish1” thus becomes the marine analog of Man1: that which is seen, conserved, and celebrated. The clownfish, the parrotfish, the seahorse–those who grace conservation posters and funding proposals–exist within a constructed ecological imaginary that privileges recognition over relationality.

In contrast, “Fish2” represents the unseen: the detritivores, the microbial life, the blennies and gobies who maintain the very breath of coral ecosystems but whose lives fall outside the data horizon. Like Wynter’s “Man2,” they are rendered infra-natural–present, but unrecognized within systems of value, modeling, and ethical consideration. The system has not been trained to see them, because human knowledge systems–shaped by extraction, utility, and dominance–have not made them legible. This echoes Stephen Cave’s argument in "The Whiteness of AI" [11], which critiques how AI often universalizes Western, white cultural assumptions as if they are neutral or objective–an erasure that maps seamlessly onto the logics of who, or what, deserves to be seen and preserved.

This ethical problem is not merely one of bias, as Bender et al. caution [12]. It is a question of paradigm. What does it mean to “conserve” when the tools of conservation are built on extractive modes of knowing? What are the ethical consequences of installing surveillance systems–even under the name of preservation–that could themselves become instruments of harm, through data misuse, ecosystem disruption, or facilitating exploitative human behaviors such as overfishing or tourism? As coral reefs bear the ecological scars of modernity–pollution, warming, acidification–they also become sites of ongoing epistemological violence, where modern tools attempt to “save” through the very logics that enabled their endangerment.

We thus propose a critical reframing: Fish1 and Fish2 are not merely taxonomic distinctions, but ethical and ontological categories. They allow us to question whose lives are rendered valuable, whose presence is made visible, and whose relations are severed by AI systems designed under the dominion of Man1 logics. Fish2 urges us to move toward a decolonial AI–one that does not presume to “know” or “save” from a distance but instead learns to be in relation with that which escapes categorization. In this way, the ethical issues of Marine-Detect call for more than algorithmic fairness or data balance; they demand a shift in how we conceptualize life, value, and the very grounds of intelligence.

By attending to these broader ethical entanglements, we seek to destabilize the dominant mode of AI deployment in ecological contexts. Rather than reinforcing systems of recognition, surveillance, and control, we must cultivate epistemologies that allow for opacity, humility, and ecological mutuality. Only then can AI move from an instrument of extraction to a companion in preservation, one that does not reproduce the violence of the Human-as-Man1 but instead imagines otherwise.

IX. Conclusion

This report has examined the Marine-Detect AI system, positioning it not merely as a technological artifact for underwater species identification but as a cognitive architecture embedded within a complex epistemic infrastructure governing marine ecological knowledge. Leveraging YOLOv8 within a PyTorch framework, the system demonstrates significant potential for automating the analysis of underwater visual data, thereby addressing critical bottlenecks in traditional marine monitoring methodologies. Our evaluation confirms the system's capability under baseline conditions but crucially reveals performance degradation under specific image alterations, particularly heavy compression and noise. This underscores a sensitivity to data quality that has direct implications for real-world deployment in variable underwater environments.

The anomalous detection patterns, such as the misclassification of noise as sea cucumbers, highlight the system's limitations and the inherent risks of relying solely on automated outputs without critical oversight. These technical findings resonate with the broader discussion of Marine-Detect as an instantiation of bounded, modular cognition. As analyzed through the lens of the Common Model of Cognition and situated cognition, the system doesn't just perceive the marine environment; it actively constructs a version of it based on its training, architecture, and the socio-technical context from which it emerges. It operationalizes a specific, technologically mediated view of marine biodiversity, necessitating a critical engagement with its 'political ontology' – how its design reflects and potentially reinforces paradigms of conservation that prioritize certain species or ecosystems based on pre-existing datasets and human-defined values.

Future analysis work must therefore transcend mere technical optimization. Addressing the limitations identified requires contending with both the 'internal knowledge system' of the model and the 'outer environment' that shapes its training and interpretation. Key directions include:

1. Enhancing Robustness: Targeted retraining or data augmentation strategies incorporating examples of common image degradations (blur, noise, compression, variable lighting) are essential to improve real-world applicability.
2. Dataset Scrutiny and Refinement: Critically evaluating the composition of the training data is paramount. This involves addressing taxonomic biases, moving beyond overly broad categories like 'fish,' and potentially incorporating data reflecting a wider range of species and ecological contexts, aligning with the more inclusive 'Fish2' framework discussed. Documenting dataset provenance and limitations transparently is crucial.
3. Investigating Anomalies: Further analysis is needed to understand the root cause of false positive patterns (e.g., sea cucumber misclassifications) and develop mitigation strategies, potentially involving architectural adjustments or refined post-processing logic.
4. Ethical Framework Integration: Moving beyond acknowledging ethical issues to actively integrating principles of fairness, accountability, and transparency into the design lifecycle. This involves considering the potential impacts of monitoring on marine life and ensuring data governance frameworks prevent misuse, engaging with the sustainability concerns raised by Bender et al.

In conclusion, Marine-Detect represents a significant step towards leveraging AI for ecological monitoring, but its

deployment compels a deeper reckoning with the intersection of technology, knowledge production, and ethics. As systems like these become increasingly integral to environmental governance, we must continually grapple with the contexts from which they are 'enunciated and created.' This means attending to the biases embedded within data and algorithms, challenging the human-centric (or 'Fish1') perspectives often encoded within them, and striving for AI systems that support a more equitable, decolonized, and ecologically representative understanding of the marine world. Adequately concluding this report requires acknowledging that the true potential of AI in conservation lies not just in its efficiency, but in its capacity to be developed and deployed responsibly, critically, and in service of a genuinely holistic ecological ethic.

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